

# Homework 1 Statistical Machine Learning

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## Problem 1

In this problem we are given two one-armed bandits  $B_1$  and  $B_2$ .  $B_1$  is run  $n_1$  times with rewards  $R_1^1, \dots, R_1^{n_1}$  and  $B_2$  is run  $n_2$  times with rewards  $R_2^1, \dots, R_2^{n_2}$ . We let the data be represented by  $X = \{R_1^1, \dots, R_1^{n_1}, R_2^1, \dots, R_2^{n_2}\}$ . We have the action space  $\aleph = \{0, 1\}$  and let  $Y = (R_1, R_2)$ , and loss functions  $L_1(Y, a) = -R_a$  and  $L_2(Y, a) = 1_{R_a=0}$ .

To find the formal Bayes rules  $\delta_i(x)$  we aim to minimise the Bayesian posterior risk function:  $\min_{a \in \aleph} \mathbb{E}[L_i(Y, \delta(x) = a) | X = x]$ ,  $i = 1, 2$ . This is quite straight forward and we get the following:

$$\delta_1(x) = \arg \min_{a \in \aleph} \mathbb{E}[-R_a | X = x] = \begin{cases} 1, & \text{if } \mathbb{E}[R_1 | X = x] \geq \mathbb{E}[R_2 | X = x], \\ 2, & \text{else,} \end{cases}$$

$$\delta_2(x) = \arg \min_{a \in \aleph} \mathbb{E}[R_a = 0 | X = x] = \begin{cases} 1, & \text{if } 1 \cdot \Pr(R_1 = 0 | X = x) \leq 1 \cdot \Pr(R_2 = 0 | X = x), \\ 2, & \text{else.} \end{cases}$$

## Problem 2

We are given the following distributions:

$$\begin{aligned} Z_a^i | \theta_i &\sim \text{Ber}(\theta_a), \quad Z_a^i = 1_{\{R_a^i=0\}}, \quad R_a^j | (Z_a^j = 1, \lambda_a) \sim \text{Exp}(\lambda_a), \\ \theta_a &\sim \text{Beta}(\alpha, \beta), \quad \lambda_a \sim \Gamma(\gamma, \delta), \quad \Pr(R_a \neq 0) = \theta_a. \end{aligned}$$

as well as that all data are iid given the parameters. To find the formal Bayes rule from problem 1 as a function of  $x$ ,  $\gamma$ ,  $\beta$ ,  $\alpha$  and  $\delta$  one can make use of the predictive posterior formula  $p(Y|X) = \int p(Y|\theta) p(\theta|X) d\theta$ , thus, the predictive posterior of this problem can be described as

$$p_{R_a|X}(r|x) = \int p_{R_a|\theta_a, \lambda_a, Z_a=1, X}(r|\theta_a, \lambda_a, Z_a = 1, x) p_{\theta_a, \lambda_a, Z_a=1|X}(\theta_a, \lambda_a, Z_a = 1|x) d\theta_a d\lambda_a.$$

We can begin by notice that

$$\mathbb{E}[R_a|X = x] = \mathbb{E}[R_a|X = x, Z_a = 1] \Pr(Z_a = 1) + \mathbb{E}[R_a|X = x, Z_a = 0] \Pr(Z_a = 0) = \mathbb{E}[R_a|X = x, Z_a = 1]\theta_a,$$

assuming the rewards cannot be negative. Thus we only need to care about when  $Z_a = 1$ . Here  $Z_a$  and  $R_a$  are to be understood as  $Z_a^{n_a+1}$  and  $R_a^{n_a+1}$ . This together with the data being iid given the parameters allows us to drop  $\theta_a$  as well as  $X$  in  $p_{R_a|\theta_a, \lambda_a, Z_a=1, X}(r|\theta_a, \lambda_a, Z_a = 1, x)$ .

Due to iid of data and that  $\lambda_a$  only affect the distribution of  $R_a$  given  $Z_a = 1$  we can write the second density in the predictive posterior as

$$p_{\theta_a, \lambda_a, Z_a=1|X}(\theta_a, \lambda_a, Z_a = 1|x) = p_{Z_a=1|\theta_a, \lambda_a, X}(Z_a = 1|\theta_a, \lambda_a, x) p_{\theta_a, \lambda|X}(\theta_a, \lambda|x) = p_{Z_a=1|\theta_a}(Z_a = 1|\theta_a) p_{\theta_a|X}(\theta_a|x) p_{\lambda_a|X}(\lambda_a|x).$$

Now, the predictive posterior can be described as

$$p_{R_a|X}(r|x) = \int p_{R_a|Z_a=1, \lambda_a}(r|Z_a = 1, \lambda_a) p_{Z_a=1|\theta_a}(Z_a = 1|\theta_a) p_{\theta_a|X}(\theta_a|x) p_{\lambda_a|X}(\lambda_a|x) d\theta_a d\lambda_a.$$

Hence, the unknowns to be found are the densities of  $\theta_a|X$  and  $\lambda_a|X$ . Beginning with  $\theta_a|X$ :

Letting  $Z_a^{1:n_a}$  be the vector of  $Z_a^1, \dots, Z_a^{n_a}$  and recalling data is iid given the parameters we have the posterior

$$p_{\theta_a|Z_a^{1:n_a}}(\theta_a|Z_a^{1:n_a}) \propto p_{Z_a^{1:n_a}|\theta_a}(Z_a^{1:n_a}|\theta_a) p_{\theta_a}(\theta_a) = \prod_i^{n_a} p_{Z_a^i|\theta_a}(Z_a^i|\theta_a) p_{\theta_a}(\theta_a) \propto \prod_{i=1}^{n_a} \theta_a^{Z_a^i} (1 - \theta_a)^{1-Z_a^i} \theta_a^{\alpha-1} (1 - \theta_a)^{\beta-1} = \theta_a^{\sum_{i=1}^{n_a} Z_a^i + \alpha - 1} (1 - \theta_a)^{n_a - \sum_{i=1}^{n_a} Z_a^i + \beta - 1},$$

which can be recognised being proportional to a beta distribution. Letting  $S_a = \sum_{i=1}^{n_a} Z_a^i$  and since  $Z_a$  is merely an indicator of the data  $X$ , we have

$$\theta_a|X \sim \text{Beta}_{\theta_a}(S_a + \alpha, n_a - S_a + \beta).$$

For  $\lambda_a|X$ : We do the same here. For the likelihood,  $X|\lambda_a = X|Z_a^{1:n_a} = 1, \lambda_a$  we have to consider that we only care for when the reward is positive (and that summing zero rewards is zero), hence,  $p_{X|Z_a^{1:n_a}=1, \lambda_a} = \lambda_a^{S_a} \prod_{i=1}^{n_a} e^{-\lambda_a R_a^i} = \lambda_a^{S_a} e^{-\lambda_a \sum_{i=1}^{n_a} R_a^i}$ . Thus,  $\lambda_a|X$  is distributed by

$$p_{\lambda|X}(\lambda|x) \propto p_{X|\lambda}(x|\lambda) p_{\lambda}(\lambda) = \lambda_a^{\gamma-1+S_a} e^{-\lambda_a(\delta + \sum_{i=1}^{n_a} R_a^i)},$$

which can be recognised as proportional to a gamma distribution

$$\lambda_a|X \sim \Gamma_{\lambda_a}\left(\gamma + S_a, \delta + \sum_{i=1}^{n_a} R_a^i\right).$$

We now have enough to compute the predictive posterior

$$\begin{aligned}
p_{R_a|X}(r|x) &= \\
\int \lambda_a e^{-\lambda_a R_a} \theta_a^{Z_a} (1 - \theta_a)^{1-Z_a} \theta_a^{S_a+\alpha-1} (1 - \theta_a)^{n_a-S_a+\beta-1} \lambda_a^{\gamma-1+S_a} e^{-\lambda_a(\delta+\sum_{i=1}^{n_a} R_a^i)} d\theta_a d\lambda_a &= \\
\int_0^\infty \lambda_a^{\gamma+S_a} e^{-\lambda_a(R_a+\delta+\sum_{i=1}^{n_a} R_a^i)} d\lambda_a \int_0^1 \theta_a^{Z_a+S_a+\alpha-1} (1 - \theta_a)^{n_a+1-Z_a-S_a+\beta-1} d\theta_a &= \\
\frac{\Gamma(\gamma + S_a + 1)}{(R_a + \delta + \sum_{i=1}^{n_a} R_a^i)^{\gamma+S_a+1}} B(Z_a + S_a + \alpha, n_a + 1 - Z_a - S_a + \beta), &
\end{aligned}$$

using that  $B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt$  and  $\Gamma(k) = \int_0^\infty t^{k-1} e^{-t} dt$ . Thus, we get the formal Bayes rule  $\delta_1(x)$  as

$$\begin{aligned}
\mathbb{E}[R_a|X = x, Z_a = 1] \theta_a &= \int R_a p_{R_a|X}(r|x) \theta_a dR_a \\
\theta_a \Gamma(\gamma + S_a + 1) B(Z_a + S_a + \alpha, n_a + 1 - Z_a - S_a + \beta) \int_0^\infty \frac{R_a}{(R_a + \delta + \sum_{i=1}^{n_a} R_a^i)^{\gamma+S_a+1}} dR_a &.
\end{aligned}$$

Letting  $C = \delta + \sum_{i=1}^{n_a} R_a^i$  and  $k = \gamma + S_a + 1$ , the integration part becomes

$$\begin{aligned}
\int_0^\infty \frac{R_a}{(R_a + C)^k} dR_a &= \int_C^\infty \frac{u - C}{u^k} du = \left[ \frac{u^{2-k}}{(2-k)} + C \frac{u^{1-k}}{k-1} \right]_C^\infty = \\
&\begin{cases} -C^{2-k}, & \text{if } S_a + \gamma > 1, \\ \infty, & \text{else.} \end{cases}
\end{aligned}$$

Thus,

$$\delta_1(x) = \arg \min_{a \in \mathbb{N}} \theta_a \Gamma(\gamma + S_a + a) B(Z_a + S_a + \alpha, n_a + 1 - Z_a - S_a + \beta) \left( -(\delta + \sum_{i=1}^{n_a} R_a^i)^{1-(\gamma+S_a)} \right).$$

For  $\delta_2(x)$  it is quite straight forward since we got

$$\delta_2(x) = \arg \min_{a \in \mathbb{N}} \mathbb{E}[R_a = 0|X = x] = \begin{cases} 1, & \text{if } \Pr(R_1 = 0|X = x) \leq \Pr(R_2 = 0|X = x), \\ 2, & \text{else.} \end{cases}$$

Since,

$$\begin{aligned}
\Pr(R_a = 0|X = x) &= 1 - \Pr(Z_a = 1|X = x) = 1 - \int p_{Z_a=1|\Theta}(Z_a = 1|x) p_\Theta(\theta) d\theta = \\
&1 - \int \theta_a p_\Theta(\theta) d\theta = 1 - \mathbb{E}[\theta_a|X = x],
\end{aligned}$$

and that we from previous computations got  $\theta_a|X \sim \text{Beta}_{\theta_a}(S_a + \alpha, n_a - S_a + \beta)$ , we get

$$\Pr(R_a = 0|X = x) = 1 - \frac{S_a + \alpha}{S_a + \alpha + n_a - S_a + \beta} = 1 - \frac{S_a + \alpha}{\alpha + n_a + \beta}.$$

Thus,

$$\delta_2(x) = \arg \max_{a \in \mathbb{N}} \frac{S_a + \alpha}{\alpha + n_a + \beta}.$$

### Problem 3

Here we are given data  $X = \{X_i\}_{i=1}^n$ ,  $X_i \sim \text{Ber}(\theta)$  conditionally independent given  $\Theta = \theta$ , a prior on  $\Theta$  such that  $p_{\Theta}(\theta) \propto \frac{e^{\theta}}{e-1} \cdot \mathbf{1}\{0 \leq \theta \leq 1\}$ , and the statistics  $n_1 = \sum_{i=1}^n X_i$  and  $n_0 = n - n_1$ .

To make a Laplace approximation  $q_{\Theta|X}(\theta|X) \sim \mathcal{N}(\theta^{MAP}, A^{-1})$  of the posterior,  $p_{\Theta|X}$ , where  $A = \frac{\partial^2}{\partial \theta^2} l(\theta)|_{\theta=\theta^{MAP}}$  we begin by finding maximum a posteriori estimation  $\theta^{MAP}$ . We have

$$p_{\Theta|X}(\theta | x) \propto p_X(x | \theta)p_{\Theta}(\theta) \propto \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i} \frac{e^{\theta}}{e-1} \cdot \mathbf{1}\{0 \leq \theta \leq 1\} \propto \theta^{n_1}(1-\theta)^{n_0} e^{\theta} \cdot \mathbf{1}\{0 \leq \theta \leq 1\},$$

hence, a negative log likelihood

$$l(\theta) = -\log p_{\Theta|X}(\theta | x) = -\log(\theta^{n_1}(1-\theta)^{n_0} e^{\theta}) + \text{constant} = -\log p_{\Theta|X}(\theta | x) = -n_0 \log(1-\theta) - n_1 \log \theta - \theta + \text{constant}, \text{ for } \theta \in [0, 1].$$

Since maximising the posterior is equivalent to minimising  $l(\theta)$  we find the zeros of the differential

$$\frac{\partial}{\partial \theta} l(\theta) = -\frac{n_1}{\theta} + \frac{n_0}{1-\theta} - 1 = 0 \text{ and } \theta \in [0, 1]$$

$$\text{and } (n_1, n_0) = (3, 7) \Rightarrow \theta^{MAP} = \frac{\sqrt{93} - 9}{2}.$$

Now to find  $A$  we have

$$A = \frac{\partial^2}{\partial \theta^2} l(\theta)|_{\theta=\theta^{MAP}} = \frac{n_1}{(\theta^{MAP})^2} + \frac{n_0}{(1-\theta^{MAP})^2} = \frac{12}{(-9 + \sqrt{93})^2} + \frac{28}{(11 - \sqrt{93})^2} \Rightarrow A^{-1} = \frac{155 - 16\sqrt{93}}{31}.$$

Thus, the Laplace approximation gives

$$\Theta|X \sim \mathcal{N}\left(\frac{\sqrt{93} - 9}{2}, \frac{155 - 16\sqrt{93}}{31}\right).$$

By scaling the true posterior with  $\frac{1}{\int p_X(x|\theta) p_\Theta(\theta) d\theta}$  and plotting the Laplace posterior approximation together with this, one can see it is a quite good approximation. Furthermore, it is even a better approximation than the Hamiltonian Monte Carlo sampling, which possibly could be explained by the true posterior being close to symmetric around its mode.

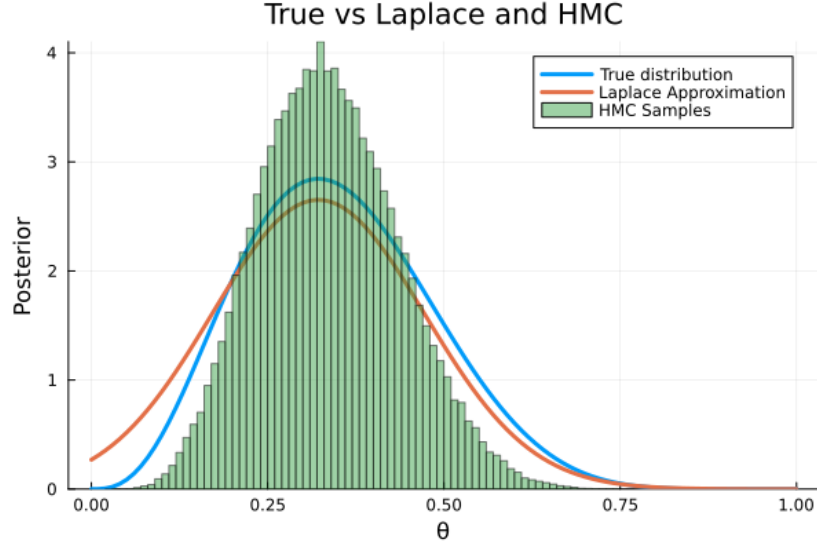


Figure 1: Plot of the true posterior together with Laplace posterior approximation and Hamiltonian Monte Carlo approximation with an acceptance rate 81% using  $10^5$  samples and leapfrog step size  $\epsilon = 0.05$ , together with the Laplace posterior approximation.